



# Accelerating Agentic and Generative AI with SambaNova

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*Director Public Sector Accounts*

**NAIRR Annual Meeting 2026**  
**10 March 2026, Arlington VA**

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- 1 Background
  - 2 Architecture Overview
  - 3 On-Premises and Cloud Deployments
  - 4 AI Use Cases

# Who We Are

## Snapshot

- Founded in 2017 by industry luminaries and originated at Stanford University
- Fully integrated generative AI platform, from 4th generation hardware to pre-trained models
- \$1.35B+ funding raised
- We are over approx. 400 employees with offices in Palo-Alto (HQ), Austin, London, Bangalore, Tokyo, Singapore, Australia

## Founded by pioneers in AI



**Clyde Hosein**  
*President & COO*



**Rodrigo Liang**  
*Co-founder & CEO*



**Kunle Olukotun**  
*Co-founder &  
Chief Technologist  
& Stanford Professor*



**Christopher Ré**  
*Co-founder &  
Stanford Professor*



**Lip-Bu Tan**  
*Chairman*

## Sophisticated, long-term investors

**BlackRock**  
Capital Investment Corporation™

**SoftBank**  
Partners Fund

**TEMASEK**

**G/**

**intel**  
Capital

**SAMSUNG**  
CATALYST  
FUND

**GIC**

**Micron**

**SK** telecom

**WALDEN**  
INTERNATIONAL

**Celesta**  
Capital

# LLM Model Trends

## Models are getting bigger

- Llama 2 70B (2023)
- Llama 3 405B (2024)
- DeepSeek R1 671B (2025)
- Kimi K2.5 1.04T (2026)
- DeepSeek R2 (*rumor* – 1.2T)

## Context Lengths are getting bigger

- Models were 4K just a few years ago
- Then 128K became the standard
- Now context lengths are growing to over 1M

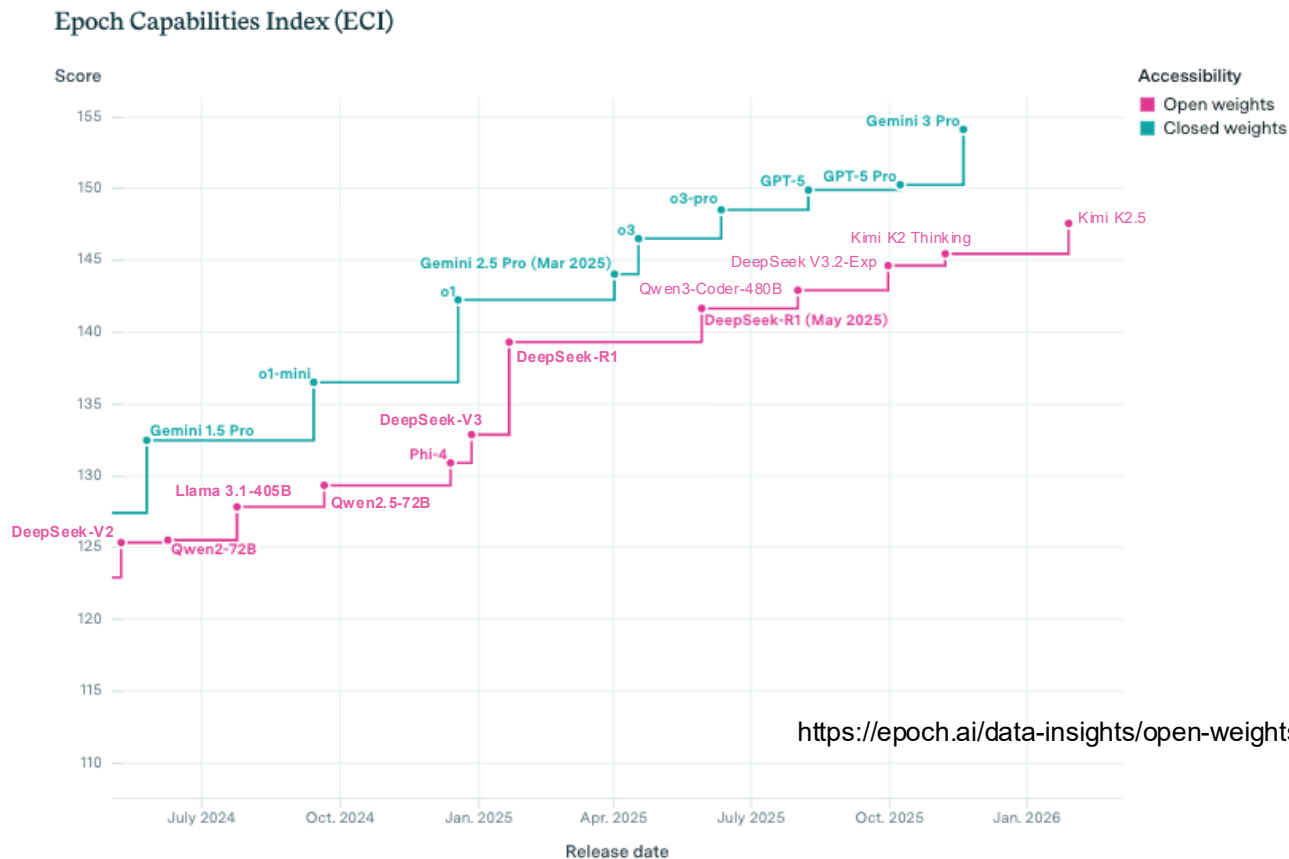
## Models are getting sparser

- Dense models were the norm (Llama 2 & 3 / GPT3)
- Then coarse Mixture of Experts (Mixtral / GPT4 both around 16 experts)
- Then fine grained Mixture of Experts (DeepSeek / Llama 4 both around 128+ experts)

## Reasoning is getting unlocked by Test Time Compute

- All the best models today are reasoning models (DeepSeek R1, OpenAI o3)
- These models generate many more tokens, which means token generation speed is more important than ever

# Open-weight models lag state-of-the-art by 3 months on average.



# Market: Journey to Agentic AI Systems

Agentic AI increase inference calls per query by 100x

Key



Unique Model



Unique Step

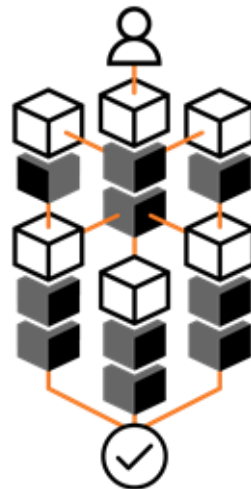
Simple Chatbot



Reasoning Models



Agentic System



- Agentic AI can increase inference calls per query by 100x

|                                 |    |     |            |
|---------------------------------|----|-----|------------|
| Tokens generated per query      | 1X | 10X | Up to 100X |
| Models Used per query           | 1  | 1   | ~5-10      |
| GPUs consume too many resources |    |     |            |

# Shallow Agents vs Deep Agents

## Shallow Agents

- Single LLM call + tool execution
- Short context windows
- Simple ReAct loops
- Quick, bounded tasks
- Limited planning capability

Example: "Search the web and summarize results"

## Deep Agents

- Extended reasoning chains
- Sub-agent orchestration
- File system access & persistence
- Hour-long autonomous workflows
- Sophisticated planning & reflection

Example: "Refactor this module and update all tests"

**Coding agents are the most successful deep agents—let's learn from their architecture**

# The 5 Pillars of Reliable Agents

Extracted from studying how the best coding agents actually work

**1**

## Orchestration

Control flow and task decomposition—how agents break down complex problems and coordinate execution

**2**

## Memory

Context management and persistence—what agents remember and how they use it to make better decisions

**3**

## Tools

External capabilities and integrations—how agents interact with systems, APIs, and data sources

**4**

## Evaluation

Testing and quality assurance—how you measure performance and continuously improve agent behavior

**5**

## Agent Skills

Composable capabilities and reusable behaviors—packaged actions that let agents perform specialized tasks reliably across domains

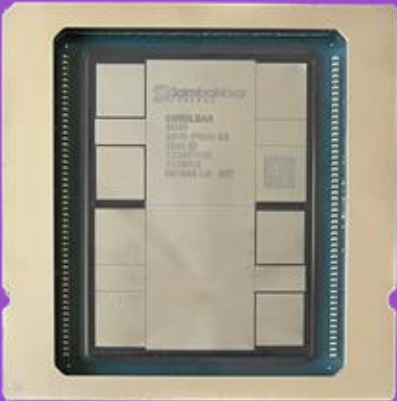


# SN40L: SambaNova's 4th Gen AI Chip

Reconfigurable Dataflow Unit (RDU)

Native multi-tenancy support with fast model switching

Ideal for production inference, multi-tenancy, agentic workflows



sambanova  
SN40L RDU

3-tier Dataflow Memory

520 MB On-Chip  
SRAM Memory

Very fast memory for high speed inference with caching

64 GB High  
Bandwidth Memory

Switch between models in as little as 2 milliseconds

768 GB High Capacity  
DDR Memory

Hold large number of models in memory

# SN40L: Tile Architecture

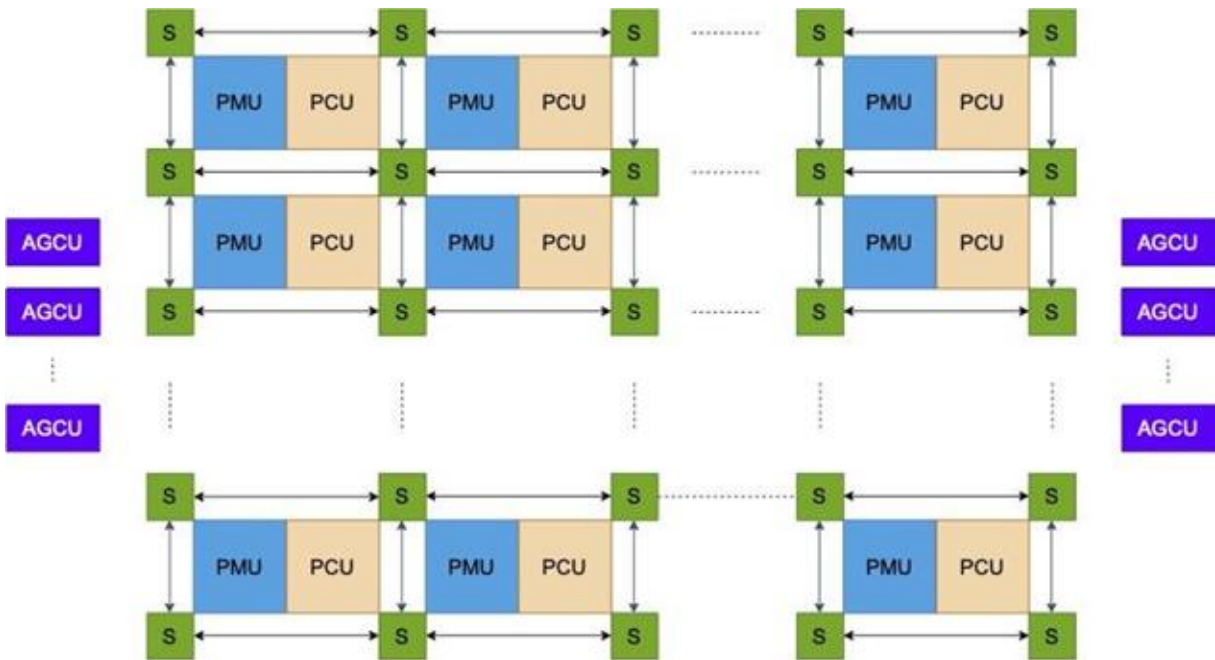
1040 PCUs and PMUs

PCU: Compute unit

PMU: Memory unit

S: Mesh switches

AGCU: Portal to off-chip memory and IO



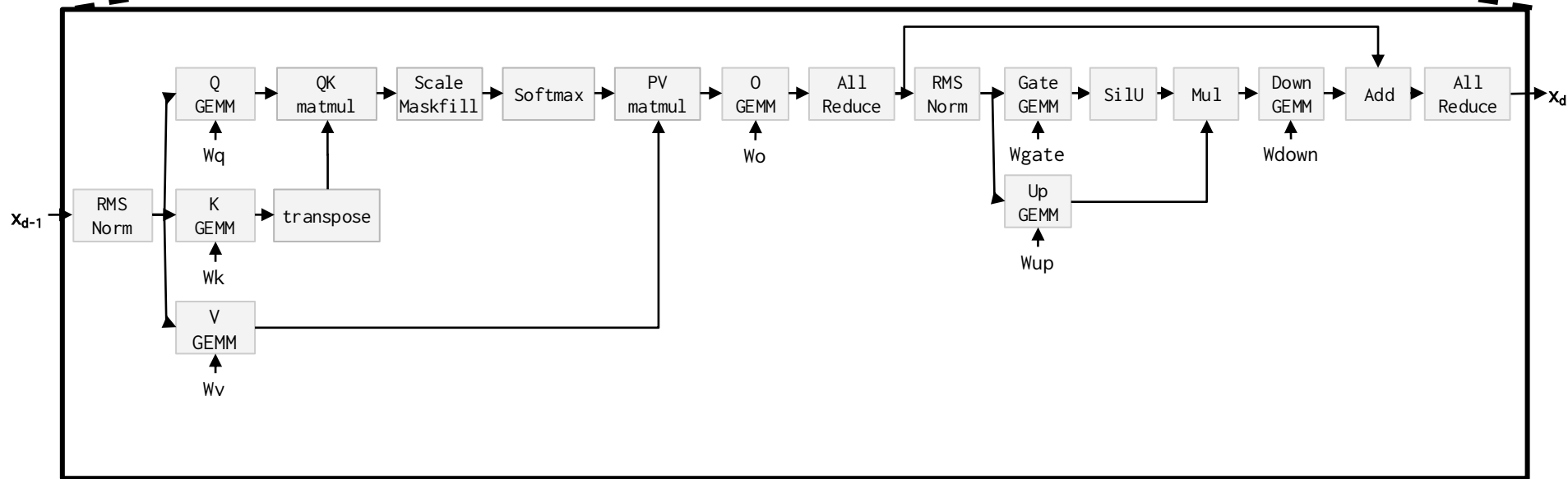
# Structure of a Transformer

Example: Llama3.1 8B



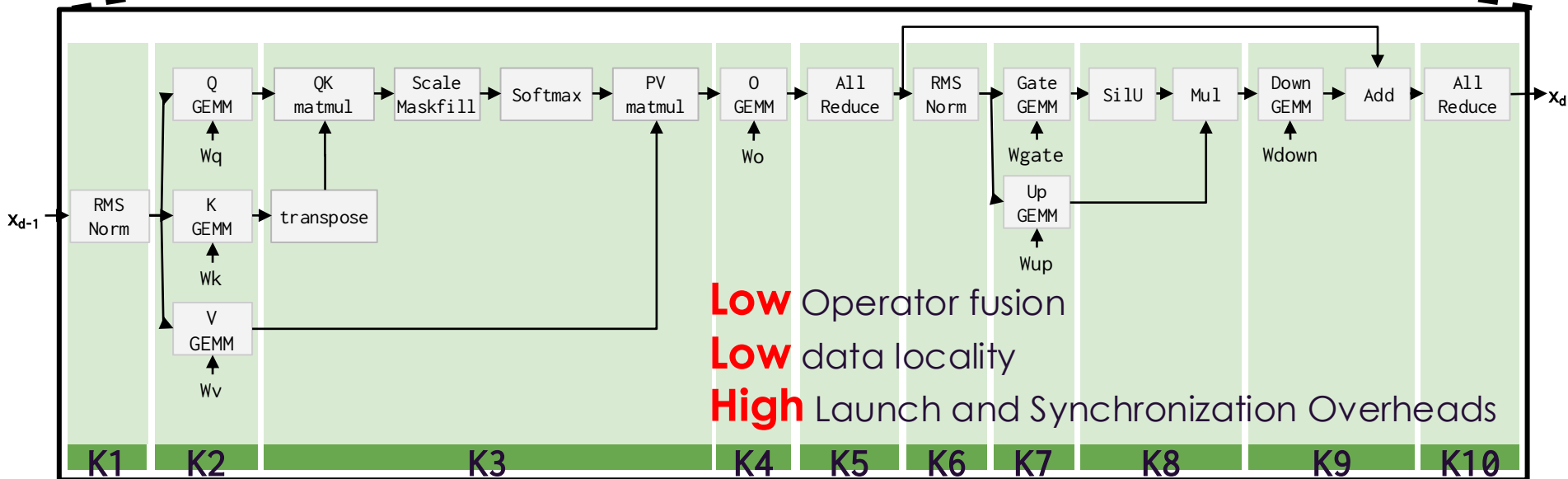
# Structure of a Transformer

Example: Llama3.1 8B



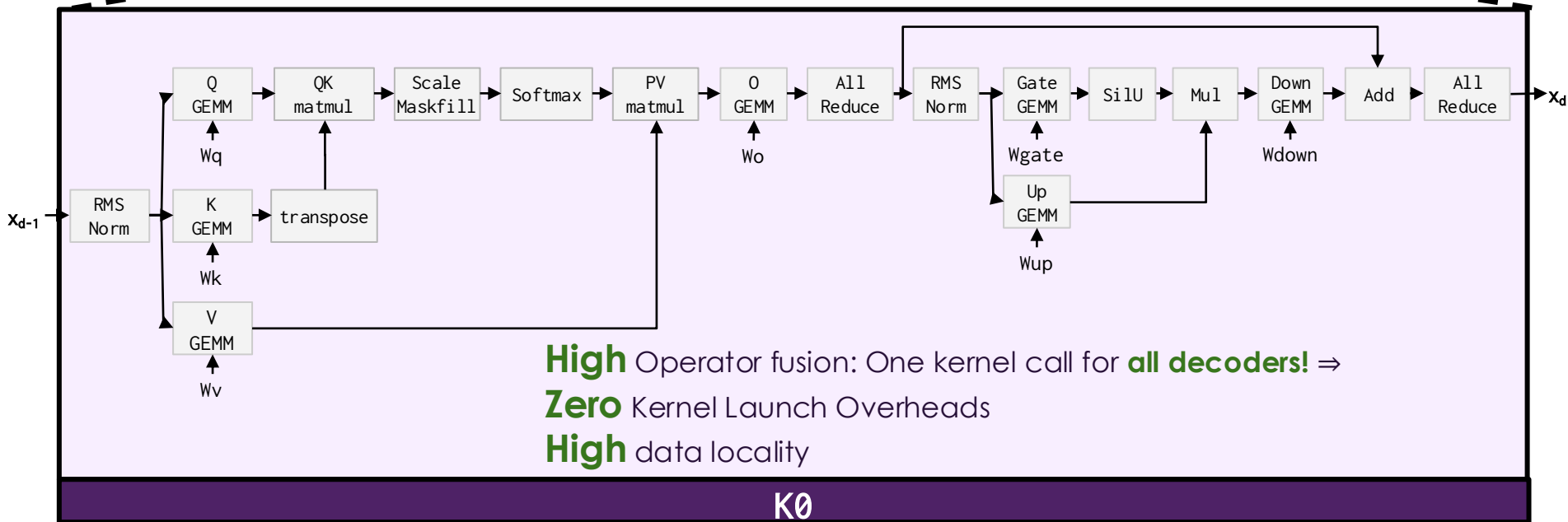
# GPUs Incur Overhead with Low Operator Fusion

Example: Llama3.1 8B



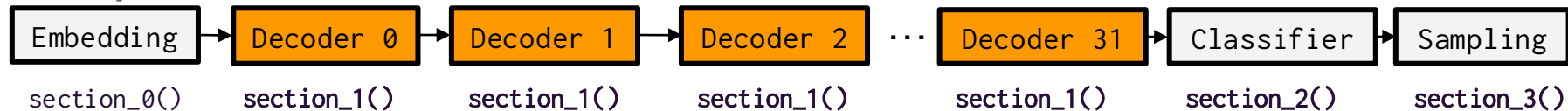
# SN40L RDU Fuses the Entire Decoder into One Kernel!

Example: Llama3.1 8B



# RDU Hardware Graph Orchestration: Temporal Fusion

## Example: Llama3.1 8B



Single Section Decoder  
+  
Repeated Section Calls  
=  
**New fusion opportunities!**

```
section_1(arg):
    Foo;
    Bar;

// Repeat 32 times
section_1(arg0)
section_1(arg1)
...
section_1(arg31)
```

**Kernel Looping**

```
section_1(args[]):
    for i in range(32):
        Foo;
        Bar;

// 1 time
section_1(args)

1 kernel launch for all decoders!
```

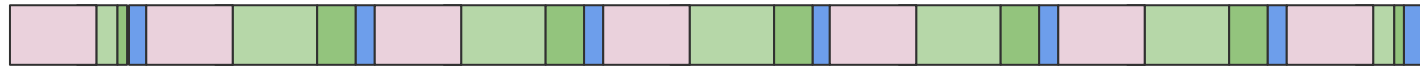
# Executing Transformer on RDU

Example: Llama3.1 8B

Baseline: **100** tokens/s



+ Single Section Decoder

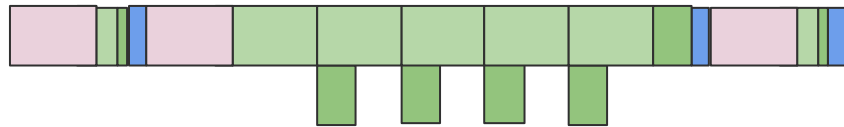


=

**500** tokens/s  
16 SN40L



+ Kernel Looping



=

**1170** tokens/s  
16 SN40L

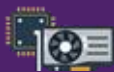




# The SN50 and SambaRack™ SN50

Shipping Q4 FY 27

# SambaNova SN50 Snapshot



**Hardware  
Features**



**Speed AND  
Throughput**

Dataflow Architecture

Agentic Caching

Cloud Scale

**5X Compute**

Vs SN40

**5X Max Speed**

Vs Nvidia Blackwell

**4X Network BW**

Vs SN40

**3X Throughput**

For Agentic Inference



## Solving AI's Data Movement Problem

# SN50 Approach to Agent Inference

The Best Balance of Cost and Speed

Gen speed VS Gen Throughput - Llama 3.3 70B, FP8, 1K, 1K



# SN50 Approach to Agent Inference

The Best Balance of Cost and Speed

Gen speed VS Gen Throughput - GPT-OSS-120B, FP8, 1K, 1K



# Full Stack AI Inference Platform

## Products:

- SambaStack – On Perm – you manage
- SambaManaged – On Prem - we manage
- SambaCloud – Our Cloud



Model Bundle, Spec Decoding, BYOC



Fully Integrated Rack-Level System  
SambaRack



SambaNova's AI Accelerator Chip  
Reconfigurable Dataflow Unit (RDU)



## Performance

Optimized for inference

## Sustainability

Most efficient energy consumption

## Efficiency

10x fewer racks/chips than the competition

## Inference Optimized

Supports custom checkpoints

## Scalability Optimized

The building block for AI clouds and supercomputers

## Rack Optimized

Comprehensive rack-level solution

# SambaStack

Inference on SambaStack has a variety of features to serve your various use cases:

## Model Bundling:

- Combine multiple models in a single OpenAI API compatible inference endpoint
- Run all models simultaneously, indicating the target model in each request

## Speculative Decoding Pairs:

- Combine a main model with a draft model in a single endpoint to increase inference speeds

## Bring Your Own Checkpoint (BYOC):

- Import external model checkpoints and run inference on them
- Imported model checkpoints can be externally fine-tuned or base models from Hugging Face

## Self-Managed Through Standard Open-Source Tools:

- Kubernetes v 1.30+, Helm v 3.8.0+, and others

### Model Bundling



### Speculative Decoding Pairs



### Bring Your Own Checkpoint

BYOC

# SambaNova SN40

Faster Inference on the Best Open Source Models



|                       | SambaNova | Nvidia* |
|-----------------------|-----------|---------|
| Llama 3.1 8B          | 932       | 180     |
| Llama 4 Maverick      | 695       | 165     |
| OpenAI's gpt-oss 120B | 743       | 321     |
| DeepSeek V3-0324      | 282       | 101     |
| DeepSeek R1-0528      | 171       | 95      |

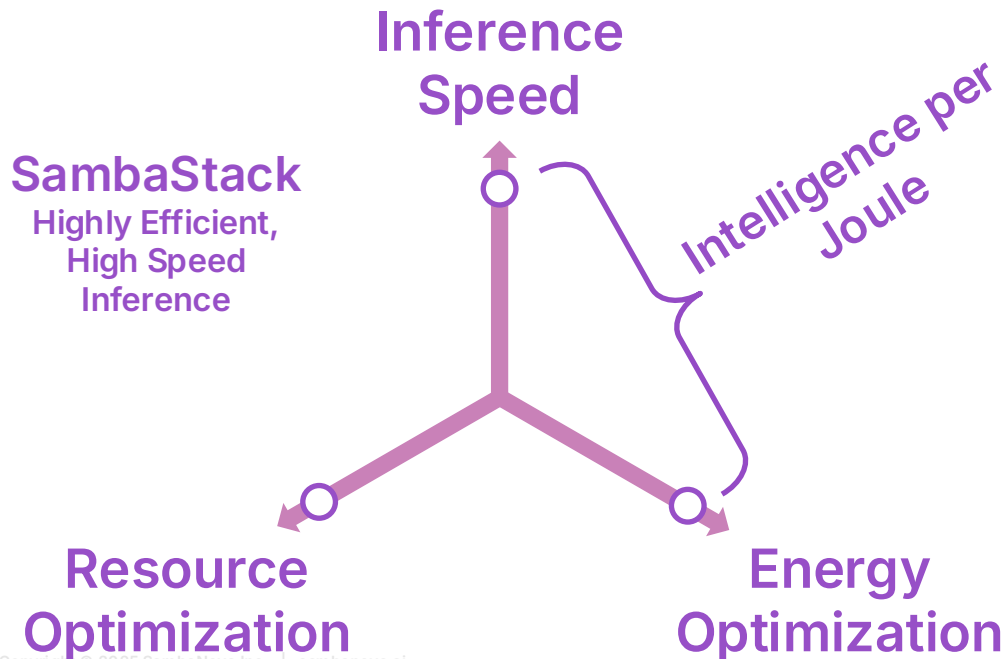
 Artificial Analysis

\*Speeds benchmarked on Nvidia GPUs on Azure

Fast inference isn't just about speed -  
it unlocks new types of applications and capabilities

# SambaNova's SambaStack

Native multi-tenancy support with fast model switching  
Ideal for production inference, multi-tenancy, agentic workflows





# The Performance Measurement that Matters

## Performance Metrics

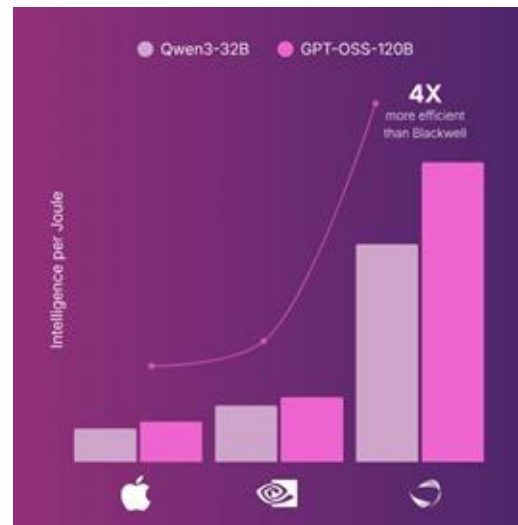
**Tokens per second:** Legacy metric that measures performance, but not quality of intelligence

**Intelligence per joule:** New metric that quantifies the end-to-end efficiency from a user's perspective, including the time cost of generation.

Provides a holistic view of the output of the entire AI system.

Measures the intelligence of the model, the efficiency of the software stack, and the energy of the hardware

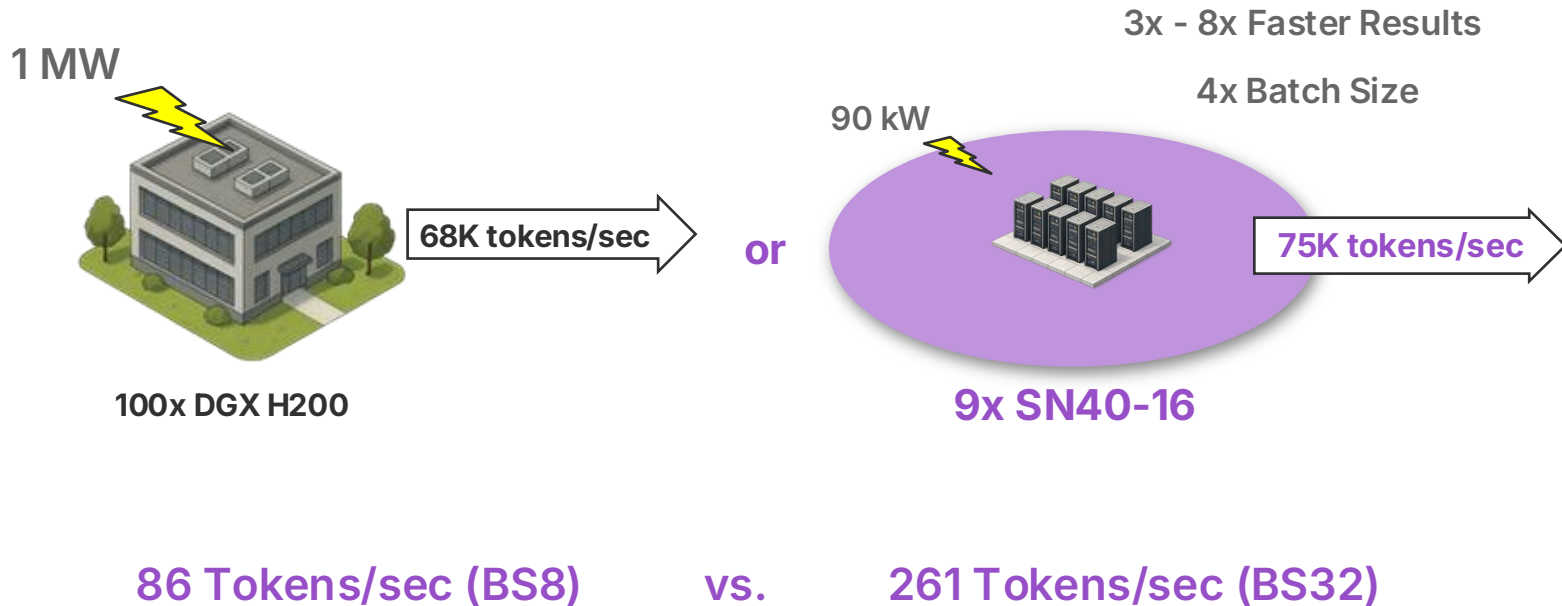
SambaNova SN40L is **4X** more efficient than B200



<https://arxiv.org/abs/2511.0788>

# The Efficient AI Factory

## Inference Performance (e.g. Llama 3.3-70B)



# The 1 Megawatt Agentic Science Factory

Inference Performance (e.g. Llama 3.3-70B)

1 MW

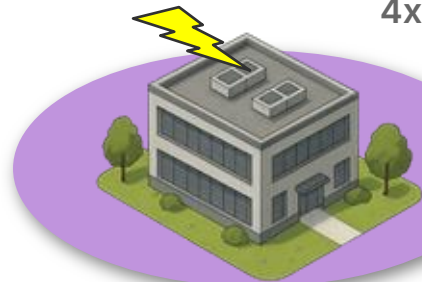


100x DGX H200

68K tokens/sec

or

1 MW



100x SN40-16

836K tokens/sec

3x - 8x Faster Results

4x Batch Size

86 Tokens/sec (BS8)

vs.

261 Tokens/sec (BS32)

# A Better Architecture for Agents

Avg 10 kW

Model Hot-Swapping

Fast  
Tokens/Second/User



## Example Agents

Java Agent

Python Agent

SW Testing  
Agent

Cloud Ops  
Agent

Knowledge  
Retrieval Agent

Contract Writing  
Agent

Travel Booking  
Agent

## Dev Frameworks



LangChain



## SambaStack Endpoint

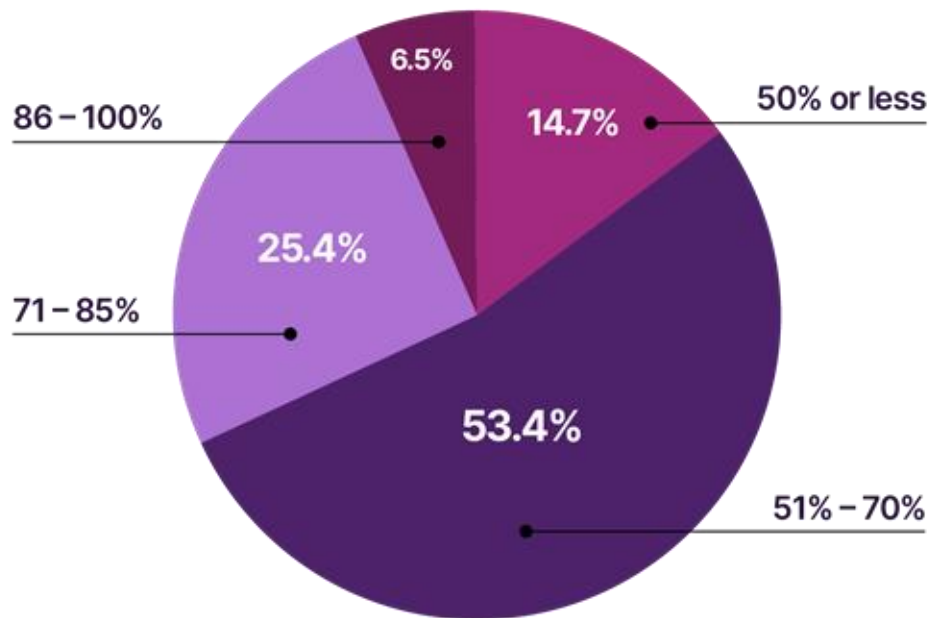
gpt-oss  
120b

DeepSeek  
V3-0324

Whisper  
v3-Large

Meta  
Llama 3.3  
70B

# More than 70% of organizations have underutilized GPUs



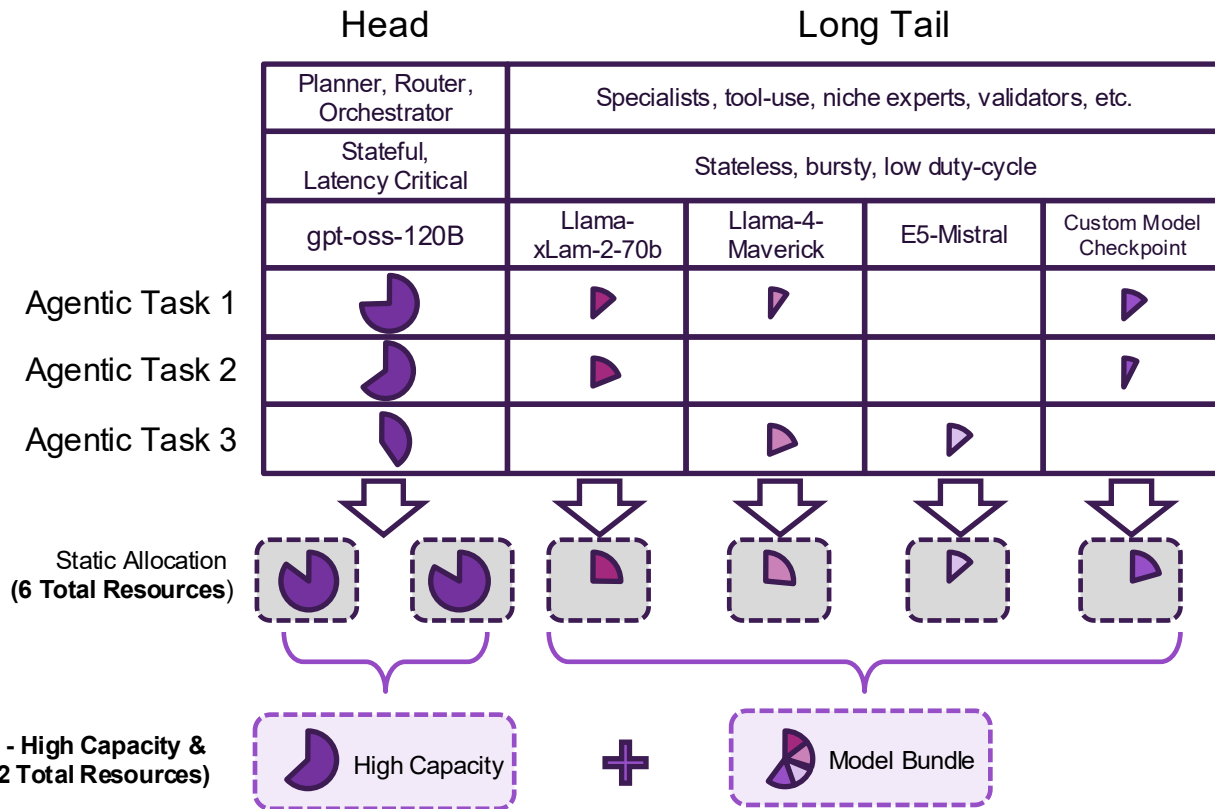
## State of AI Infrastructure at Scale Survey

**15%** of respondents report that fewer than 50% of available GPUs are in use

**53%** of respondents believe less than 70% of their GPUs are utilized.

**A better solution is required**

# Model Bundles – Agentic Workloads



# SambaCloud

# SambaCloud

Open source models available today



Llama 4-Maverick  
Llama-3.3-70B-Instruct  
Llama-3.3-Swallow-70B-Instruct-v0.4  
Llama-3.1-8B-Instruct



Qwen3

Qwen3-235B  
Qwen3-32B



deepseek

DeepSeek-R1 0528  
DeepSeek-V3.1  
DeepSeek-V3.2  
DeepSeek-V3.1-Terminus  
DeepSeek V3-0324  
DeepSeek-R1-Distill-Llama-70B



OpenAI

GPT-OSS-120B  
Whisper Large v3



Mistral E5 7B

HUMAIN

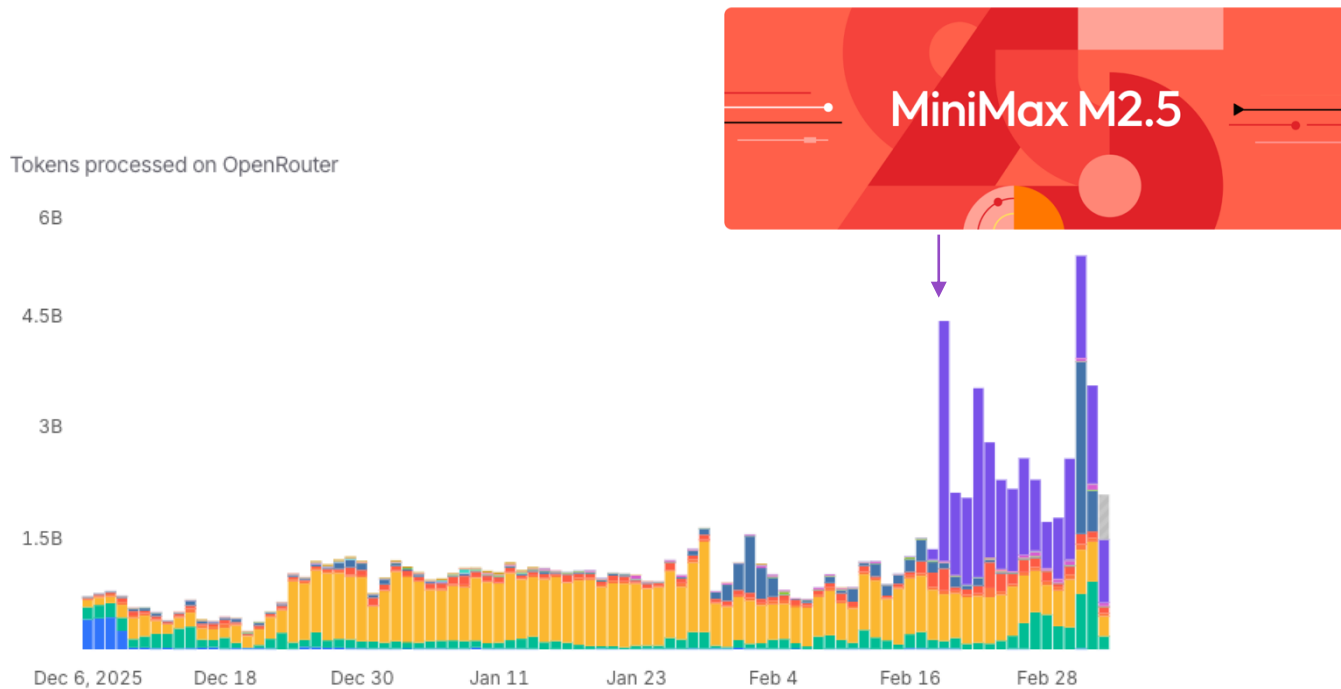
ALLaM-7B

SambaNova is delivering the largest and most capable open source models, with unparalleled performance to unlock new capabilities that have been impossible to achieve

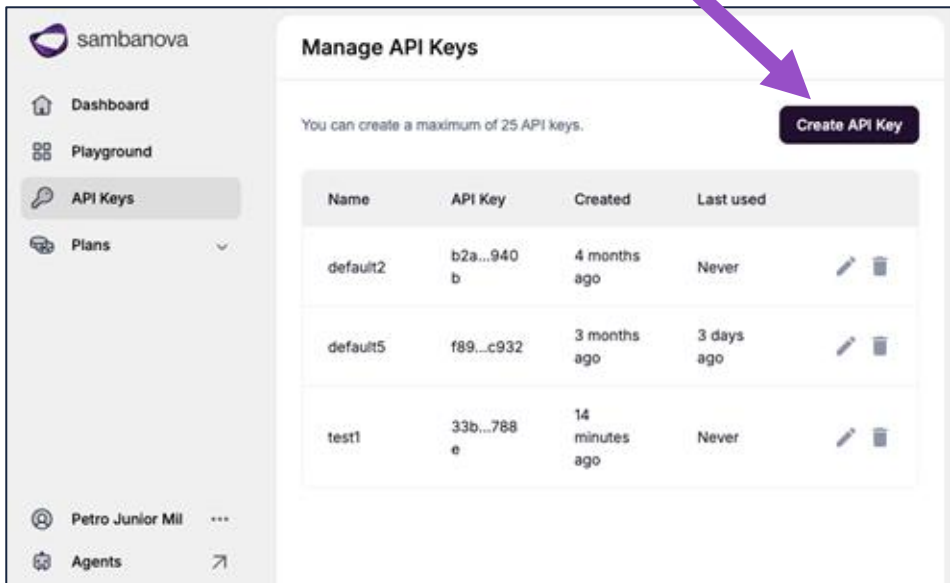


# SambaCloud

Newest Model – MiniMax M2.5



# Get Your API Key at [cloud.sambanova.ai](https://cloud.sambanova.ai)



**Manage API Keys**

You can create a maximum of 25 API keys.

[Create API Key](#)

| Name     | API Key    | Created        | Last used  |                                                                                                                                                                     |
|----------|------------|----------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| default2 | b2a...940b | 4 months ago   | Never      |   |
| default5 | f89...c932 | 3 months ago   | 3 days ago |   |
| test1    | 33b...788e | 14 minutes ago | Never      |   |



```
hello-world.js  hello_world.py

from sambanova import SanbaNova

client = SanbaNova(
    base_url="your-sambanova-base-url",
    api_key="your-sambanova-api-key"
)

completion = client.chat.completions.create(
    model="Meta-Llama-3.3-70B-Instruct",
    messages = [
        {"role": "system", "content": "Answer the question in a couple sentences."},
        {"role": "user", "content": "Share a happy story with me"}
    ]
)

print(completion.choices[0].message.content)
```


sambanova

Dashboard
Playground
API Keys
Plans

Login/Signup \$5 credit
Agents
AI Starter Kits
Community
Documentation
Connect with us

## AI Starter Kits

Developer Community View all starter kits on GitHub

Starter Kits Help You Build Fast – They bootstrap application development for common AI use cases with open-source Python code on SambaNova GitHub. They let you see how the code works, and customize it to your needs, so you can prove the business value of AI.

### QuickStart - Cloud ReadMe

How to get started with SambaNova Cloud – A comprehensive guide to help you begin your journey with SambaNova Cloud.

Getting Started

### Multi Modal Retriever

Chart, Image, and Figure Understanding – Unlock insights from complex PDFs and images with advanced retrieval and answer generation that combines both visual and textual data.

Advanced AI Capabilities Demo

### SambaAI Workspaces Integration

Seamless LLM Integration in Google Workspace – Enhance your productivity by integrating powerful language models directly into Google Docs and Sheets via App Scripts.

Advanced AI Capabilities

### Llama 3.1 Instruct-o1

Enhanced Reasoning with Llama 3.1 405B – Experience advanced thinking capabilities with Llama 3.1 Instruct-o1, hosted on Hugging Face Spaces.

Advanced AI Capabilities Demo

### Function Calling

Tools calling implementation and generic function calling module – Enhance your AI applications with powerful function calling capabilities.

Advanced AI Capabilities Demo

### Enterprise Knowledge Retrieval

Document Q&A on PDF, TXT, DOC, and more – Bootstrap your document Q&A application with this sample implementation of a Retrieval Augmented Generation semantic search workflow using the SambaNova platform, built with Python and a Streamlit UI.

Intelligent Information Retrieval Demo

### Search Assistant

Include web search results in responses – Expand your application's knowledge with this implementation of the semantic search workflow and prompt construction strategies, with configurable integrations with multiple SERP APIs.

Intelligent Information Retrieval Demo

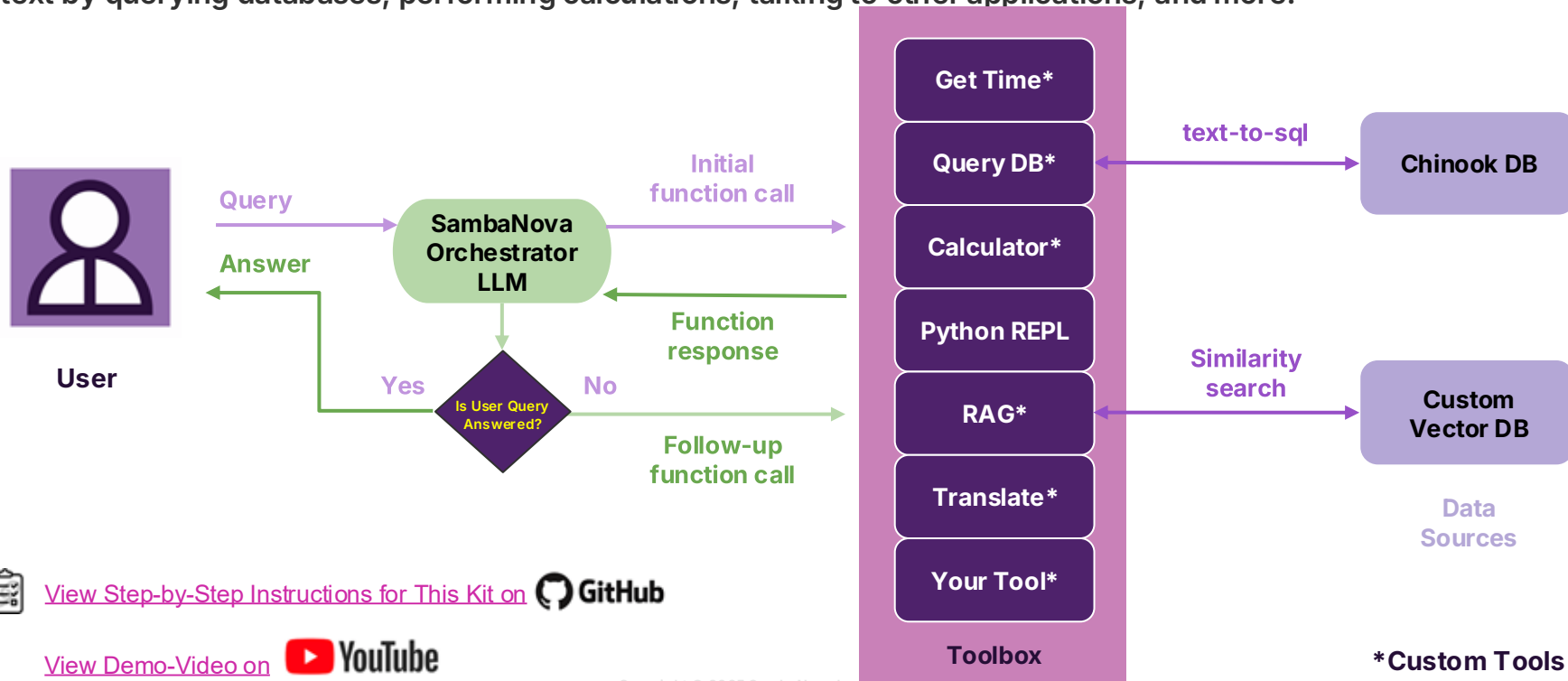
### Benchmarking

Compare model performance – Quickly determine which models meet your speed and quality needs by comparing model outputs, Time to First Token, End-to-End Latency, Throughput, Latency, and more with configuration options in a chat interface.

Model Development & Optimization Demo

# Tool/Function Calling

Easily enable business use cases by adding function calls into NLP applications that do more than understand text by querying databases, performing calculations, talking to other applications, and more.



[View Step-by-Step Instructions for This Kit on](#)  **GitHub**

[View Demo-Video on](#)  **YouTube**

# Agentic Applications



## Input Objective

User provides high-level goals



## Planning

Agent decomposes goals into tasks  
1 Large LLM



## Execution

Agent performs or delegates tasks  
Many Small LLMs



## Feedback and Learning

System improves through iteration

✓ Reduces manual intervention

✓ Increases efficiency and speed

✓ Enables complex multi-step task automation

[Agents Demo \(Try it yourself!\)](#)

# SambaNova Integrations Ecosystem

55+ integrations across agents, frameworks, tools, and platforms — [docs.sambanova.ai](https://docs.sambanova.ai)

### Agent Frameworks

 LangGraph

 CrewAI

 AutoGen

 Google ADK

 AG

 CA

 SK

 CO

 A0

 Semantic Kernel

 Composio

 Agent Zero

 Docker

### Coding Assistants

 Claude Code

 Cursor

 Continue

 Cline

 RC

 KC

 VS Code

 QW Code

 BlackBoxAI

 OpenClaw

 Google WS

### LLM Frameworks

 LangChain

 LlamaIndex

 Haystack

 LiteLLM

 Hugging Face

 Llama Stack

 Vercel

 Gradio

 Oumi

 AI Suite

### Real-time Voice

 ElevenLabs

 LiveKit

 Pinecone

 HumeAI

 Vapi

### Low-code Platforms

 DI

 FW

 Langflow

 n8n

 Make

 DataRobot

### Eval & Monitoring

 IN

 LM Eval

 Weave (W&B)

### Tools & Search

 BU

 FM

 IS

 Linkup

 Open WebUI

### DB, Cloud & Video

 Milvus

 Neo4j

 MO

 AWS

 TwelveLabs

**55+** Integrations    
 **10** Agent Frameworks    
 **11** Coding Assistants    
 **10** LLM Frameworks    
 **5** Voice Platforms    
 **6** Low-code

# THANK YOU

[tim.clarke@sambanova.ai](mailto:tim.clarke@sambanova.ai)

# Try It for free!

<https://cloud.sambanova.ai>



## Papers about SN40L

<https://arxiv.org/abs/2405.07518>



*"SambaNova SN40L:  
Scaling the AI Memory  
Wall with Dataflow and  
Composition of Experts",  
MICRO '24*

<https://arxiv.org/abs/2410.23668>



*"Kernel Looping:  
Eliminating Synchronization  
Boundaries for Peak  
Inference Performance"*